
What students need to learn

3.1 Short-term preparation

Short-term physiological preparation

Warm-up:

Stages, types of stretching, intensity, duration

Sources of energy for exercise:

Students need to understand energy systems (the ATP-PC system, Lactic and aerobic pathways and the energy continuum) and the following sources of energy for exercise:

- dietary manipulation
- carbo-loading
- hydration
- creatine loading

Short-term acclimatisation:

Environmental factors that influence preparation and competition, heat adaptation, hydration planning, increase in plasma volume, increased sweat rate, effects of altitude

Short-term psychological preparation

Motivation and stress control:

- anxiety (cognitive and somatic, state and trait)
- theories of anxiety and arousal (Drive Theory, inverted-U Hypothesis, Catastrophe Theory, zone of optimum performance)
- effects on technique, eg choking, aggression versus assertion

Strategies:

Mental rehearsal, use of visualisation and imagery, 'self talk', pre-game routines, relaxation techniques, somatic and cognitive techniques including PMR (Progressive Muscle Relaxation), control of aggression

External influences:

Home advantage, crowd effect, social facilitation, importance of competition, evaluation apprehension, strategies for coping, environmental factors

Short-term technical preparation

Kit and equipment:

Selection, environmental and cultural factors, effects on performance

Ergogenic aids for short-term preparation:

Drugs/supplements, clothing (compression clothing, ice vests), cycle ergo meters for warm-up

Use of holding camps and pre-match rituals

- The differences between short-term holding camps and long-term preparation camps
- Use of pre-match rituals in preparation for optimum performance

Fatigue and the recovery process

Fatigue:

Effects on performance, depletion of fuels (PC, glycogen), waste product accumulation (lactic acid), central governor theory, dehydration, electrolyte loss

Recovery:

Time phases (first few hours — 24 hours)

- First few hours — cooling down, lactic acid removal, restoration of ATP-PC and glycogen stores, EPOC (excess post-exercise oxygen consumption)
- 24 hours — DOMS (delayed onset muscle soreness), hydration, carbohydrate loading

Use of ergogenic aids — ice baths, compression clothing, music